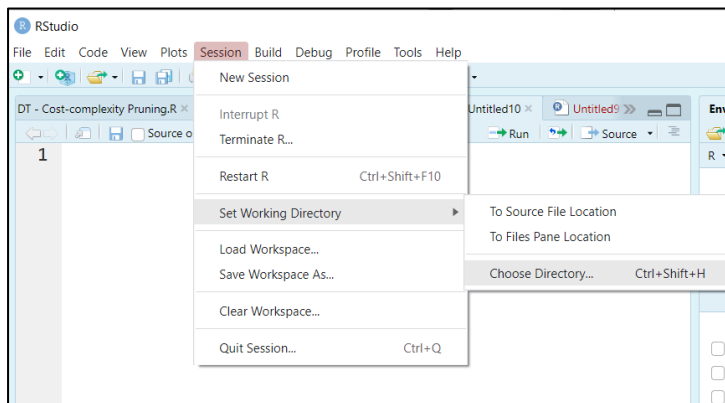


Pictorial Week Four : Logistic Regression and Model Building

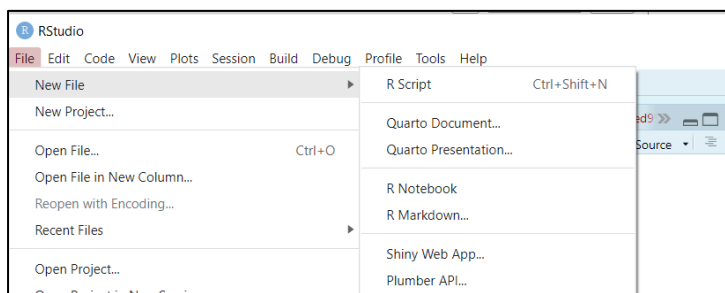
How to: Set Working Directory

Session > Set Working Directory > Choose Directory (Pick the folder where you saved the downloaded dataset)



How to: Open A New R Script

File > New File > R Script



How to: Import Dataset

The Housing Data.csv dataset is used to demonstrate.

```
housing_data = read.csv("Housing Data.csv")
```

How to: Apply the glm() function

The Housing Data.csv dataset is used to demonstrate.

```
## Defining a new binary variable near_bay, coded as 1 when the ocean proximity is classified as NEAR BAY, else 0 (don't need to know)
near_bay = ifelse(housing_data$ocean_proximity == "NEAR BAY", 1, 0)

## Fitting a logistic regression model
ObjectName = glm(near_bay ~ total_bedrooms + median_income + median_house_value, family = "binomial", data = housing_data)
```

On the Cheat Sheet

20	<code>glm(y ~ x1 + x2 + x3)</code>	A function used to fit generalized linear models. We will use it in this course to fit logistic regression models e.g. <code>mod1 <- glm(y ~ x1 + x2 + x3, family = "binomial")</code>
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How to: Apply the summary() function

```
summary(ObjectName)|
```

How to: Apply the as.factor() function

The Housing Data.csv dataset is used to demonstrate.

```
## Converting Numeric variables to factor variables  
total_bedrooms = as.factor(housing_data$total_bedrooms)
```

How to: Apply the lm() function

The Example Data.csv dataset is used to demonstrate.

```
example_data = read.csv("Example Data.csv")  
  
## Fitting a linear regression model  
ObjectName = lm (Y ~ X, data = example_data)
```

On the Cheat Sheet

26	<code>lm(y ~ x1 + x2 + x3)</code>	Fits a multiple linear regression model of y as a function of variables x1, x2 and x3
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How to: Apply the step() function

```
step.model <- step(fit.empty, scope = list(lower = fit.empty, upper = fit.full), direction = 'forward')
```

On the Cheat Sheet

44	<code>step(fit2,direction="forward", scope=list(upper=fit1,lower=fit2))</code>	Perform stepwise regression e.g. For forward selection in a dataset called fresh. <code>fit1 <- lm(demand ~ ., data=fresh); fit2 <- lm(demand ~ 1, data=fresh); step(fit2,direction="forward", scope=list(upper=fit1,lower=fit2))</code>
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